



Environmental risk and ecotoxicological assessment of postcoagulation water treatment sludge for the reclamation of anthropogenically degraded soils

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Abstract

Purpose This study aims to investigate the physicochemical and ecotoxicological properties of postcoagulation water treatment sludge (PCWTS) in the context of its use for anthropogenic land reclamation. It also attempts to promote the beneficial reuse of post-coagulation sludge as a valuable resource, with the intention of contributing to future research in this field.

Materials and methods In the course of the research, PCWTS after the water treatment process by a prehydrolysed coagulant, polyaluminium chloride (PAC, $[\text{Al}_2(\text{OH})_n\text{Cl}_{6-n}]_m$), and soil from three anthropogenically altered areas were subjected to in-depth physicochemical (UV–VIS, F-AAS, WDXRF, CHNS, and SEM–EDS) and ecotoxicological analyses and assessments. The potential ecological risk index (PERI), potential ecological risk factor (ER^i), and geoaccumulation index (I_{geo}) were calculated and interpreted. Statistical analyses (ANOVA and Tukey's HSD test) were also used to further clarify the ecotoxicological results.

Results and discussion The ecological risk assessment of heavy metal pollution indicated low-to-moderate risk levels ($\text{ER}^i < 80$ and $\text{PERI} = 206$), according to the background of the soils of the Upper Silesia region. The calculated values of ER^i were in the order of $\text{Ni} > \text{Cu} > \text{Cd} > \text{Cr} > \text{Pb} > \text{Zn}$, whereas I_{geo} was $\text{Ni} > \text{Cu} > \text{Cr} > \text{Zn} > \text{Cd} > \text{Pb}$. The results of the ecotoxicological tests demonstrated the potential suitability of the PCWTS as an amendment for anthropogenically degraded soils. The use of PCWTS as a soil-improving agent does not increase the risk of heavy metal contamination in soils classified as suitable for use in both agricultural and green areas, and contributes to improved soil fertility by providing essential nutrients (N 9.7 g/kg, P 2.3 g/kg, K 22.9 g/kg).

Conclusion Despite the presence of minor heavy metals, PCWTS applied to degraded soils does not pose a high risk of HMs, resulting in nontoxicity and beneficial effects on plant growth. The use of PCWTS for land reclamation in industrially degraded areas is an example of sustainable sludge management in water treatment plants and aptly corresponds with the ideas of circular economy and the principles of sustainable development. While this study provides valuable short-term insights based on a 14-day evaluation, the absence of a long-term assessment remains a limitation that warrants further investigation.

Keywords Water treatment sludge · Aluminium coagulants · Heavy metals · Ecotoxicity · Risk evaluation · Circular economy

Abbreviations

WTP	Water treatment plant	HMs	Heavy metals
WTS	Water treatment sludge	ER^i	Potential ecological risk factor
PCWTS	Postcoagulation water treatment sludge	PERI	Potential ecological risk index
WWTS	Wastewater treatment sludge	I_{geo}	Geoaccumulation index
		PAC	Polyaluminium chloride

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ACH	Aluminium chlorohydrate
COD	Chemical oxygen demand
TOC	Total organic carbon
DOC	Dissolved organic carbon
EC	Electrical conductivity
TSS	Total suspended solids
TDS	Total dissolved solids
SEM	Scanning electron microscopy
EDS	Energy dispersive spectroscopy

1 Introduction

The use of advanced surface water treatment processes is associated with the production of varying amounts of waste in the form of sediments, which may contain various pollutants. For this reason, they should be stored in landfills or disposed of in accordance with all the relevant environmental regulations. However, the sludge originating from water treatment processes can also be treated or beneficially reused in line with the principles of green chemistry, circular economy and sustainable development.

Water treatment requires the application of several key unit processes. One of the stages is coagulation, in which organic and inorganic compounds are removed in suspended, colloidal and dissolved forms. The most commonly used coagulants in the water treatment process (WTP) are nonhydrolysed aluminium salts. These include aluminium sulfate (VI) ($\text{Al}_2(\text{SO}_4)_3$, alum) and aluminium chloride (AlCl_3) (Pająk 2023; Łukasiewicz 2016; Tan et al. 2024). In many cases, aluminium sulfate (VI) is the first-choice coagulant because it is inexpensive, available, safe to handle, and easy to use (Bérubé and Dorea 2008; Matilainen et al. 2005). Nevertheless, the effective removal of contaminants, production of high-quality water, and increased need to optimize the cost-to-price ratio require the use of prehydrolysed coagulants (e.g., polyaluminium chlorides, PAC, $[\text{Al}_2(\text{OH})_n\text{Cl}_{(6-n)}]_m$, and aluminium chlorohydrate, ACH, $\text{Al}_n\text{Cl}_{3n-m}(\text{OH})_m$), which display many advantages over conventional (nonhydrolysed) coagulants (Jiang and Graham 1998; Łukasiewicz and Rzaşa 2017; Nowacka et al. 2014).

There are several practical methods for recovering and utilizing the reuse of postcoagulation water treatment sludge (PCWTS). Some of them include binding of hydrogen sulfide (H_2S) in sewage treatment plants, the removal of phosphorus during wastewater treatment, the immobilization of phosphorus in eutrophic lakes (Ahmad et al. 2016), as well as its external reuse for the manufacturing of construction materials such as cement and bricks (Mohamed et al. 2022). PCWTS can be used for road surfaces, for instance in the case of soil-sludge mixtures with the addition of cement

in paving structures as sub-bases, soil-sludge mixtures with the addition of lime as sub-bases, and to strengthen the ground (Takao et al. 2024). Additionally, PCWTS can be used to improve soil properties during land reclamation, thereby protecting the surface of a closed landfill from water and wind erosion (Balkaya 2016).

Compared with wastewater treatment sludge (WWTS), the application of PCWTS for land reclamation has not been widely reported in the subject literature (Boudjabi and Chenchouni 2021; Lue-Hing 2023; Tamanini et al. 2008). However, it seems to be a promising strategy, especially in areas degraded by mining activity. The physicochemical characteristics of PCWTS allow it to be utilized as a soil substitute because of its content of essential nutrients such as nitrogen (N), phosphorous (P), potassium (K), calcium (Ca) and magnesium (Mg) (Ashworth et al. 2020; McCauley et al. 2011). However, the total concentrations of plant nutrients in aluminium-based residuals do not necessarily reflect plant availability. According to the Polish Waste Act (Parliament of the Republic of Poland 2012), sludge and backwash water from WTPs should be treated as hazardous waste because of their contents of toxic metals, inorganic and organic contaminants and harmful microorganisms. Surface water, especially river water, contains many contaminants; thus, the use of sediments could lead to recontamination of the environment (Ahmad et al. 2016). The composition and properties of PCWTS depend largely on its source, including the characteristics of the raw water and the applied treatment process. Changes in drinking water quality have been the subject of many studies for many years (Piedgoń 2024; Mensah-Akutteh et al. 2022; Ouyang et al. 2006). The quality of raw water may vary seasonally and reflect contaminant inputs resulting from activities carried out within the catchment area. Therefore, the composition of PCWTS produced even within one water treatment plant may change (Fiore et al. 2022). The properties of the PCWTS are also influenced by the type and dosage of coagulant used (aluminium- and iron-based), as well as the particular stages of the WTP, including technological processes such as thickening, dewatering or storage (Ahmad et al. 2016; Fiore et al. 2022; Płonka et al. 2026; Takao et al. 2024). For these reasons, the results obtained for sludge from a single water treatment plant should not be directly extrapolated to residuals generated at other facilities. Considering the potential environmental risks and variability associated with the WTP, the assessment proposed in this study takes into account multiple criteria, including the following three aspects.

Firstly, it needs to be noted that in the case of PCWTS, there is a risk of environmental pollution due to the presence of heavy metals (HMs), such as arsenic (As), barium (Ba), cadmium (Cd), chromium (Cr), copper (Cr), nickel (Ni),

lead (Pb) and zinc (Zn) (Bartoszewski 1996; Leszczyńska 2009). The contamination of soils by HMs due to their toxicity, long-term persistence and bioaccumulation capacity may lead to environmental problems. However, the environmental risk depends not only on total concentrations but also on HMs fractionation and speciation, soil pH, and redox changes, which affect their mobility and potential uptake by plants.

In addition, the high aluminium content of PCWTS deposited in soil requires particular attention. The solubility, speciation and bioavailability of aluminium are strongly dependent on the soil pH. Under acidic conditions ($\text{pH} < 5.5$), the amount of exchangeable and potentially toxic form of aluminium (Al^{3+}) in the soil solution increases and becomes more available for plants. (Nguyen et al. 2022; Zhao et al. 2018). Moreover, Al-based water treatment residuals have a high phosphorus sorption capacity, which may reduce its bioavailability and potentially affect plant growth (Zhao et al. 2018). Aluminium speciation may also be influenced by interactions with dissolved organic matter (Weng et al. 2022). Additionally, when assessing the environmental safety of PCWTS application to soil, potential long-term changes associated with its ageing (Agyin-Birikorang and O'Connor 2009) or weathering over time should also be considered. Because of this, it is necessary to carry out physicochemical and ecotoxicological studies to determine the composition and properties of PCWTS, and to identify its potential adverse effects on the environment, even when it is applied in degraded areas.

Secondly, phytotoxicity tests are crucial for environmental monitoring and risk assessment, especially in cases of waste disposal in contaminated soils (EMPPPO 2007). Standard parameters analysed during these studies, typically including root and shoot length, germination percentage, and biomass production for selected species of test organisms, should be determined.

Third, to assess the environmental risk associated with the potential introduction of heavy metals, the potential ecological risk factor (ER^i), the potential ecological risk index (PERI) and the geoaccumulation index (I_{geo}), should be calculated (Hakanson 1980; Muller 1969). Considering these circumstances and the lack of comprehensive and sufficient scientific work in this field, the objective of this study was to assess the short-term suitability of PCWTS containing aluminium compounds as an amendment for three anthropogenically degraded soils. The investigations carried out to assess the potential use of the PCWTS for land reclamation included: (i) physicochemical characterization of the PCWTS and degraded soils, (ii) environmental risk assessment calculations based on HMs content, (iii) short-term ecotoxicological testing, and (iv) short-term laboratory-scale pot tests. The specific novel contribution of this study

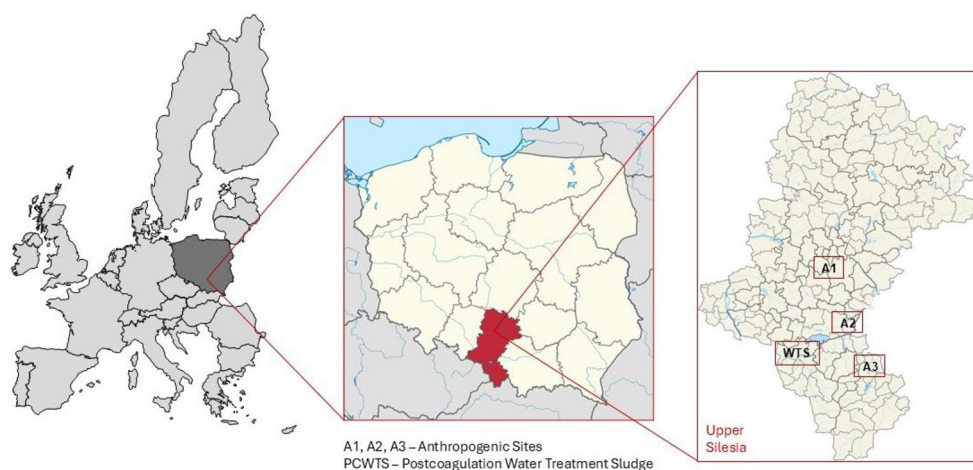
are the integrated assessment of PCWTS, which combines physicochemical characterization, ecological risk assessment, and short-term plant response in different industrially degraded soils. Although the land application of water treatment residuals has been investigated (Dayton and Basta 2001; Alvarenga et al. 2018), their use for the reclamation of anthropogenically degraded and contaminated soils, including mine spoil areas, remains relatively poorly documented. The subject of the research carried out aligns with the concept of circular economy (European Commission 2019) and the idea of sustainable development, providing an approach to the use of byproducts, minimizing, and even improving the condition of the environment.

2 Materials and methods

2.1 Postcoagulation water treatment sludge and anthropogenic sites

PCWTS samples were collected repeatedly over a three-month period at several-day intervals (24 individual samples in total) from a water treatment plant located in the Silesian Voivodeship. The samples were collected after implementing the water treatment process via a prehydrolysed coagulant, polyaluminium chlorides, PAC, and $[\text{Al}_2(\text{OH})_n\text{Cl}_{6-n}]_m$ ($\text{pH } 4.4 \pm 0.1$, $d \ 1.125 \pm 0.113 \text{ g/mL}$, $\text{Al } 11.5 \pm 0.1\%$, $\text{chloride } 6.5 \pm 0.1\%$, $\text{alkalinity } 80.1 \pm 0.1\%$). The test samples were obtained by mixing and accurately averaging individual samples. The samples from three anthropogenically transformed sites (A1, A2 and A3) were collected from areas degraded by industrial activity from the surface in the most industrialized region of Poland, the Upper Silesian Coal Basin. Two sites are located in the central part of the basin, and one is on the border of Silesia and Malopolska Voivodships. Samples A2 and A3 were collected from coal mine spoil heaps, and sample A1 was from an industrial area heavily contaminated with heavy metals. The primary samples from the anthropogenic sites were collected from depths of 0–20 cm at ten sampling points in accordance with ISO 18400-104. The collected samples were placed in plastic bags and thoroughly mixed to obtain laboratory samples. An average of 25 kg of laboratory samples of PCWTS and A1, A2 and A3 were transported to the laboratory and stored in a refrigerator at $4 \pm 1^\circ\text{C}$. Before physicochemical analysis, all the samples were air-dried at ambient temperature (14 days, $20 \pm 1^\circ\text{C}$, relative humidity, $55 \pm 5\%$), crushed in a ball mill and sifted through a 2 mm sieve (Retsch GmbH, Germany). The sludge and soil samples were analysed as described in the analytical procedures section. The locations of the sampling sites are presented in Fig. 1.

Fig. 1 Locations of PCWTS sampling points and soil samples from anthropogenic sites



2.2 Physicochemical analysis of solid samples

The moisture content and dry matter content were determined by drying the sample at 105 ± 1 °C to a constant weight according to PN-EN 12880:2004. Ash and organic matter contents were determined by burning the sample at 550 °C according to PN-EN 12879:2004. The total phosphorus (Total P) content was determined spectrophotometrically (Nanocolor® UV-VIS, Macherey-Nagel GmbH, Germany) according to PN-Z-15011-3, whereas the total nitrogen (Total N) amount was determined according to the Kjeldahl method described in PN-EN 13342:2002. The concentrations of metals (Cd, Cr, Cu, K, Ni, Pb, and Zn) in the PCWTS were determined via flame atomic absorption spectrometry (FAAS) according to PN-ISO 8288:2002 (SolaarS4, Thermo Fisher Scientific, USA) after prior mineralization of the samples in *aqua regia* (HCl+HNO₃ 3:1) under reflux cooling. The contents of Al, P, Cd, Co, Cr, Cu, K, Mn, Mo, Ni, Pb, and Zn in the soil samples were measured via wavelength-dispersive X-ray fluorescence spectrometry (WDXRF) (Rigaku ZSX Primus, Rigaku Holdings Corp., USA). The infrared spectroscopy method (ELTRA CHNS, Eltra GmbH, Germany) was used to determine the content of total sulfur (Total S). The concentrations of total dissolved solids (TDS) and salinity, as well as measurements of pH and electrical conductivity (EC) in water leachate (water extracts, soil: deionized water, 1:5), were determined via the analytical procedures described in detail in Sect. 2.3. SEM/EDS analysis was performed via a high-resolution scanning electron microscope (Apreo 2 S LoVac; Thermo Fisher Scientific, USA). The studied sample was placed on double-sided adhesive carbon tape (SPI Supplies, West Chester, USA) and then sputtered with Au (2.5 nm) under vacuum in an Ar atmosphere via a Safematic CCU-010 HV high vacuum compact coating unit (Safematic GmbH, Zizers, Switzerland). For EDS analysis, an EDS UltraDry detector was used

(Thermo Fisher Scientific, USA). T1 (backscattered electrons) and ETD (Everhart–Thornley) (secondary electrons) were used for imaging. All the experiments were performed in triplicate.

2.3 Physicochemical analysis of water leachates

The water leachate (water extract) was prepared according to PN-EN 12457-2:2006 (dry matter: deionized water 1 kg:10 L), an overhead shaker (Rotax 6.8, Velp Scientific, Italy), 24 h, 20 rpm, 21 ± 1 °C). After shaking, the water extract was separated via a vacuum filter (0.45 µm, Whatman, UK). The pH and electrical conductivity (EC) were measured electrometrically (CPC-401, Elmetron®, Poland) according to PN-EN ISO 10523:2012 and PN-EN 27888:1999, respectively. Total suspended solids (TSS) and sulfate (VI) were determined gravimetrically via methods described in PN-EN 872:2007 and PN-ISO 9280:2002, respectively. Chloride was measured via the Mohr method according to PN-ISO 9297:1994. Chemical oxygen demand (COD), total and dissolved organic carbon (TOC, DOC), total phosphorus (Total P), total nitrogen (Total N) and the concentrations of Ca²⁺, Mg²⁺, SO₄²⁻, and F⁻ were determined spectrophotometrically (Nanocolor® UV-VIS, Macherey-Nagel GmbH, Germany) via test kits (Macherey-Nagel, Germany). The concentrations of trace metals (Cd, Cr, Cu, Ni, Pb, and Zn) were quantified via FAAS (SolaarS4 AA Series Spectrometer®, Thermo Fisher Scientific®, USA) prior to digestion with *aqua regia* according to PN-ISO 8288:2002. The calibration curves for each metal were prepared via five standard solutions (Certipur®, Merck®, Germany). The R² values were no less than 0.9993 (Ni), 0.9999 (Cu), 0.9991 (Pb), 0.9999 (Cr), 0.9985 (Cd), and 0.9993 (K). A segmented calibration curve fit with a minimum curvature of -4% and a maximum curvature of 7% was used for Zn determination. All the experiments were performed in triplicate.

2.4 Microbiotests

The ecotoxicity assessment was carried out via second-generation microbiotests with test organisms (MicrobioTests Inc., Belgium), both for the solid PCWTS and their water leachates. The extracts were prepared from a sample of PCWTS (dry matter: deionized water 1 kg:10 L, overhead shaker (Rotax 6.8, Velp Scientific, Italy), 24 h, 20 rpm, 21 ± 1 °C). The solid particles were removed from the extracts via centrifugation (4000 rpm, 30 min, DM0 424, Chemland, Poland). The laboratory conditions of the microbiotests are presented in Table 1. All tests were performed in triplicate.

2.5 Pot tests

This test aimed to determine the effect of PCWTS amendment on short-term plant growth in samples collected from anthropogenic sites. For the plant test, a mixture of universal grass seeds consisting of 30% *Lolium perenne*-Aut, 10% *Festuca rubra*-Dipper, 10% *Festuca rubra*-Aido, 5% *Festuca rubra*-Capricio, 15% *Poa pratensis*-Lincolnshire, and 15% *Festuca arundinacea*-Starlett (Rolimpex SA, Itawa, Poland) was used. The study was conducted in pots measuring 6.6-6 cm with drainage holes. Before analysis, the samples from the anthropogenic sites and sludge were air-dried, crushed and sieved through a 2 mm sieve. PCWTS was added to A1, A2, and A3 at weights of 10, 15, and 20 wt% and blended. A weight of 0.5 g of grass seeds was subsequently placed at a depth of 0.5 cm. Pots were watered with 20 mL of distilled water per day, covered in the plant propagator and stored under laboratory conditions (22 ± 1 °C) with exposure to daylight for 12 h daily. After 14 days, the sprouted grass biomass was harvested, and the plants were subsequently washed under running water to remove soil particles and weighed. All the experiments were performed in triplicate.

2.6 Environmental risk assessment

2.6.1 Potential ecological risk index

The potential ecological risk index (PERI) was used to assess the ecological risk posed by HMs pollution from PCWTS into soils, surface water, and groundwater. The PERI is the sum of the individual potential ecological risk factors (ER^i) of all the HMs present in the soil and is calculated via Eq. (4) (Hakanson 1980).

$$PERI = \sum_{i=1}^n ER^i \quad (4)$$

The potential ecological risk factor (ER^i) of the sludge sample was calculated according to Eq. (5):

$$ER^i = T_r^i \times \frac{C_i}{C_b} \quad (5)$$

where T_r^i stands for the toxic response factor for potentially toxic elements, reflecting their toxicity and sensitivity in the ecosystem. The T_r^i values for the metals (mg/kg) Cd, Cu, Cr, Ni, Pb and Zn are 30, 5, 2, 5, 5, and 1, respectively (Chen et al. 2019; Gong et al. 2024; Hakanson 1980); C_i represents the measured metal concentration in the PCWTS test; and C_b is the background concentration of the HMs in the reference area. In this study, various types of C_b in reference areas were used: (i) the median values of the concentrations of HMs in Polish soils from grazing lands ($n=258$) were 0.3, 4, 5, 4, 13 and 35 for Cd, Cr, Cu, Ni, Pb and Zn, respectively (Pasiieczna and Markowski 2014), and (ii) the median values of the determined HMs in Upper Silesia soils ($n=1564$) were 1.3, 5, 7, 5, 44, and 104 for Cd, Cr, Cu, Ni, Pb and Zn, respectively (Lis and Pasiieczna 1995). The interpretation of the ecological risk index values is shown in Fig. 3.

2.6.2 Geoaccumulation index

The geoaccumulation index (I_{geo}) of the HMs was used to assess the level of metal pollution in the sludge and degraded soils. I_{geo} was calculated according to Eq. (6) (Muller 1969)

$$I_{geo} = \log_2 \frac{C_i}{1.5C_b} \quad (6)$$

where C_i represents the measured concentration of the metal in the sludge sample (mg/kg); C_b stands for the geochemical background concentration of the metal (mg/kg) in the environment for both Poland in general (Pasiieczna and Markowski 2014) and Polish Upper Silesia region specifically (Lis and Pasiieczna 1995); and the factor 1.5 is a constant value used to account for natural fluctuations in the background concentration. The evaluation standard of the I_{geo} index is shown in Fig. 4.

2.7 Statistical analysis

Statistical analysis was performed by using statistical software (Statistica™ 13.3; TIBCO, USA), and the results are shown as the mean value $\pm$ standard error (SE). Analysis of variance (ANOVA) was applied to the data from the pot tests described in Sect. 3.3. Tukey's HSD (honestly significant difference) test was used to detect statistically significant differences ($p \leq 0.05$) between the groups of samples.

Table 1 Test conditions for assessing the ecotoxicity of PCWTS to microbiotests

Microbiotest	Test organism	Test conditions	Response
Phytotoxkit® for liquid samples	<i>Lepidium sativum</i> , <i>Sinapis alba</i> L.	Test duration: 72 h Temperature: 25 °C±2 °C Control: Reconstituted water Preliminary test: 10 seeds per 10 mL of test sample Measuring of root length: a digital calliper	Inhibition (I) of root growth compared with the control sample, according to formula
Phytotoxkit® for solid samples	<i>Lepidium sativum</i> , <i>Sinapis alba</i> L.	Test duration: 72 h Temperature: 25 °C±2 °C Control: Reference soil Preliminary test: 10 seeds per test sample Volume of distilled water added to sample: 46 mL Measuring of root length: a digital calliper	$I(\%) = \frac{L_c - L_s}{L_c} \cdot 100 (1)$ L_c-average length of root in control sample L_s-average length of root in tested sample Inhibition (I) of luminescent bacteria after 15 and 30 min, according to formula:
Luminescent bacteria test kit LCK 480	<i>Aliivibrio fischeri</i>	Test duration: 2 h Temperature: 15 °C±2 °C Control: NaCl, 2% solution (20 g NaCl/L distilled water) Standard: K ₂ Cr ₂ O ₇ (solution 22.6 mg/L)	$I_t(\%) = \left(1 - \frac{I_s}{I_c}\right) \cdot 100 (2)$ I_s-measured luminescent effect of tested sample after time I_c-measured luminescent effect of control sample after time Immobilization (I_{mob}) of individuals, according to formula: $I_{mob}(\%) = \frac{N_{mob}}{N_{total}} \cdot 100 (3)$ N_{mob}-number of immobilized individuals after 48 h N_{total}-number of total individuals
Daphtoxkit F®	<i>Daphnia magna</i> S.	Hatching of the Ephippia: of min 6000 lx Test duration: 72 h Temperature: 21 °C±1 °C Control: Reconstituted water Reference substance: 0.6 mg/L of K ₂ Cr ₂ O ₇ Preliminary test: 20 daphnia per 10 mL of test sample	

3 Results

3.1 Analysis of the physicochemical parameters of the PCWTS and anthropogenic site samples

The physicochemical parameters of the PCWTS, which define its suitability for application in anthropogenic soils, and samples collected from anthropogenic sites (A1–A3) are presented in Table 2. Analysis of the sludge composition revealed dry matter and organic matter at the levels of 22.1% and 30.3%, respectively, with a total carbon content of 10.22%. The measured concentration of total nitrogen was 9.7 g/kg, while the total sulfur concentration was 3.5 g/kg.

The results of the physicochemical analysis of the A1–A3 samples revealed that the organic matter content ranged from very low (4.6%) to high (25.7%). The measured parameters responsible for the salinity at the anthropogenic sites revealed that sample A2, which was collected from a coal mine spoil heap, presented the highest values of EC (2980 $\mu\text{S}/\text{cm}$) and TDS (1493 mg/L) in comparison with samples A1 (108 $\mu\text{S}/\text{cm}$ and 53.9 mg/L, respectively) and A3 (685 $\mu\text{S}/\text{cm}$ 342 mg/L), respectively. The pH value of sample A2 was strongly acidic (pH 2.8), which corresponded to high salinity (1580 mg NaCl/L). In comparison, samples A1 and A3 were moderately acidic (pH 5.25) and alkaline (pH 8.35), while the PCWTS sample was nearly neutral (pH 6.5).

Analysis of the PCWTS sample at the soil-to-water ratio (L: S=5:1) revealed that the concentrations of TDS and salinity were 126 and 122 mg/L, respectively. The results revealed that the EC value of the sludge was 251 $\mu\text{S}/\text{cm}$, whereas the corresponding pH values after 2 and 24 h reached 6.50 and 6.55, respectively.

The concentrations of metals in the tested samples and their limit values for soils classified into Groups II–IV according to Polish regulations (Minister of the Environment, 2016)

are presented in Table 3. The total P content in the PCWTS (2.3 g/kg) was much higher than that in samples collected from anthropogenic sites (0.3–0.4 g/kg). The concentration of K in PCTWS (22.9 g/kg) was three times greater than that in A1 and slightly lower than that in A2 and A2. The content of Al in the PCWTS was 56.4 g/kg, indicating that the Al_2O_3 concentration in the sludge sample amounted to 10.7%. Samples A2 and A3 were characterized by higher concentrations of aluminium than samples A1 and PCTWS were. The concentrations of HMs in the sludge varied in the order of $\text{Co} < \text{Cd} < \text{Pb} < \text{Cr} < \text{Ni} < \text{Cu} < \text{Zn}$. The highest content was observed for Zn (184 mg/kg) and Cu (75 mg/kg), whereas the lowest amount was measured for Cd (2.2 mg/kg). The most contaminated soils, in terms of heavy metals, were found in sample A1. The concentrations of Zn, Pb, and Cd were 6797, 954, and 36 mg/kg, respectively. The highest concentrations of Cr (95 mg/kg) and Ni (58 mg/kg) were detected in sample A2, whereas the highest concentrations of Cu (51 mg/kg) and Co (23 mg/kg) were detected in sample A3. The contents of Cu (75 mg/kg) and Ni (74 mg/kg) in PCWTS were greater than those in samples collected from anthropogenic sites. Nevertheless, the results of this study indicate that the concentrations of all the HMs in the PCWTS were within the acceptable range for land application in soils classified as Groups II–IV. Research has clearly shown that the disposal of PCWTS in soils classified for use in agricultural, wooded, shrubby, and green areas may be safe and does not pose a risk of soil contamination. The results of the SEM/EDS analysis of the PCWTS are presented in Table 4.

The results indicated that the largest atomic and weight shares in the analysed PCWTS samples were O (58.9% and 51.3%), C (20.1% and 13.1%), Si (9.4% and 14.3%), Al (7.8% and 11.4%), and Fe (1.4% and 4.2%). The contents of macro- and micronutrients, including P, K, Ca, Mg, S,

Table 2 Selected physicochemical parameters of the PCWTS and anthropogenic site samples

Parameter	Unit	Results			
		PCWTS	A1	A2	A3
Total nitrogen (N_t)	g/kg	9.7 $\pm$ 1.0	n.d	n.d	n.d
Total sulfur (S_t)	g/kg	3.5 $\pm$ 0.4	n.d	n.d	n.d
Total carbon (C_t)	%	10.2 $\pm$ 1.0	n.d	n.d	n.d
Dry matter (d.m.)	%	22.9 $\pm$ 2.0	n.d	n.d	n.d
Ash (A^a)	%	69.7 $\pm$ 7.0 ^a	95.4 $\pm$ 9.5 ^a	74.3 $\pm$ 7.4 ^a	75.0 $\pm$ 7.5 ^a
Organic matter (OM)	%	30.3 $\pm$ 3.0	4.6 $\pm$ 0.5	25.7 $\pm$ 2.6	24.9 $\pm$ 2.5
Electrical conductivity (EC)	$\mu\text{S}/\text{cm}$	251 $\pm$ 13	108 $\pm$ 5	2980 $\pm$ 149	685 $\pm$ 34
Salinity	mg NaCl/L	122 $\pm$ 6	51 $\pm$ 3	1580 $\pm$ 79	335 $\pm$ 17
Total Dissolved Solids (TDS)	mg/L	126 $\pm$ 6	54 $\pm$ 3	1493 $\pm$ 75	342 $\pm$ 17
pH					
after 2 h	n.a.	6.50 $\pm$ 0.05	5.25 $\pm$ 0.05	2.80 $\pm$ 0.05	8.35 $\pm$ 0.05
after 24 h	n.a.	6.55 $\pm$ 0.05	5.30 $\pm$ 0.05	2.90 $\pm$ 0.05	8.05 $\pm$ 0.05

^a values in the analytical state

n.d. not determined

$f_{\text{TDS}}=0.5$ const

Table 3 Concentration of metals in PCWTS and anthropogenic site samples; permissible values of risk-causing substances in different groups of soils

HMs	Unit	Result	The permissible values					
		PCWTS	A1	A2	A3	Group II	Group III	Group IV
Al	g/kg	56.4 ± 3.4	15.4 ± 0.1	84.7 ± 5.1	85.2 ± 5.1	n.a.	n.a.	n.a.
K	g/kg	22.9 ± 2.3	7.1 ± 0.7	17.4 ± 1.7	17.7 ± 1.8	n.a.	n.a.	n.a.
P	g/kg	2.3 ± 0.3	0.3 ± 0.04	0.4 ± 0.1	0.4 ± 0.1	n.a.	n.a.	n.a.
Cd	mg/kg	2.2 ± 0.2	36 ± 3.6	2.0 ± 0.2	6.0 ± 0.6	2–5	10	15
Co	mg/kg	< 0.05	2.9 ± 0.3	13 ± 1	23 ± 2	20–50	100	200
Cr	mg/kg	51 ± 5	12.5 ± 1.3	95 ± 10	56 ± 6	150–500	500	1000
Cu	mg/kg	75 ± 8	43.8 ± 4.4	33 ± 3	51 ± 5	100–300	300	600
Ni	mg/kg	74 ± 7	9.7 ± 1.0	58 ± 6	26 ± 3	100–500	300	500
Pb	mg/kg	49 ± 5	954 ± 95	24 ± 2	113 ± 11	100–250	500	600
Zn	mg/kg	184 ± 18	6797 ± 678	97 ± 10	173 ± 17	300–50	1000	2000

n.a. not applicable

Group II – agricultural soils, Group III – wooded and shrub lands as well as green areas, Group IV – industrial lands and mining grounds

and Mn, ranged from 0.2 to 0.5% in the case of the atomic analysis and from 0.3 to 1.5% in the case of the weight analysis. Additionally, SEM/EDS analysis revealed that the percentage atomic and weight contents of other components, including heavy metals, did not exceed 0.1 and 0.3%, respectively. A visual interpretation of the SEM/EDS analysis results for the PCWTS sample is presented in Fig. 2.

SEM/EDS analysis of the PCWTS sample with quantitative mapping is shown in Fig. 3, which illustrates the spatial distribution of the main detected elements, including C, O, Si, Ca, Fe, Al, Na and Mg, along with trace elements

(Pb, Cd, Cr, Ni, and Zn). The chemical composition of the PCWTS indicates a predominance of mineral phases, mainly oxides (E_nO_m), aluminosilicates ($AB_xO_{2x} \cdot nH_2O$, where A: Na, K, Li, Ca, Ba, Sr, B: Si, and Al) and iron (Fe), which are typical for sediments formed during coagulation and flocculation in the water treatment process. The high carbon content suggests the presence of trace amounts of organic matter or organic residues from the treatment process. On the basis of the obtained results, the absence of potentially toxic elements in the tested range confirms the neutral, mineral nature of the sludge.

Table 4 Composition of the elements in the PCWTS according to the atomic and weight percentages via SEM/EDS

Element	Atomic content (%)		Weight content (%)	
	Result	Error	Result	Error
Na	n.d.	n.d.	n.d.	n.d.
Mg	0.5	< 0.1	0.6	< 0.1
Si	9.4	< 0.1	14.3	0.1
Cr	n.d.	n.d.	n.d.	n.d.
Ni	n.d.	n.d.	n.d.	n.d.
Cu	0.1	< 0.1	0.3	< 0.1
Zn	< 0.1	< 0.1	0.1	< 0.1
Cd	n.d.	n.d.	n.d.	n.d.
Pb	< 0.1	< 0.1	0.2	0.1
C	20.1	0.1	13.1	0.1
O	58.9	0.5	51.3	0.5
Al	7.8	< 0.1	11.4	0.1
P	0.2	< 0.1	0.3	< 0.1
S	0.2	< 0.1	0.3	< 0.1
K	0.6	< 0.1	1.4	< 0.1
Ca	0.3	< 0.1	0.7	< 0.1
Ti	0.1	< 0.1	0.2	< 0.1
Mn	0.5	< 0.1	1.5	< 0.1
Fe	1.4	< 0.1	4.2	0.1

n.d. not detected

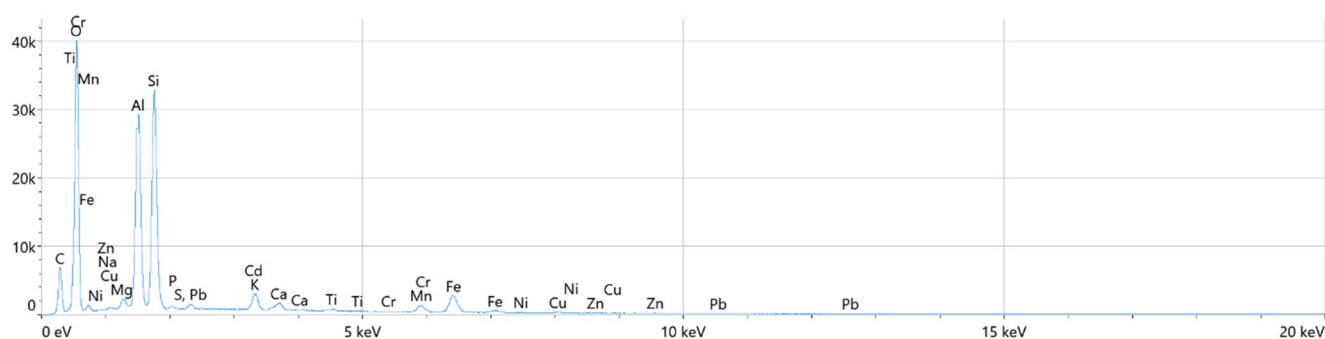


Fig. 2 Chemical analysis of the PCWTS sample via SEM/EDS

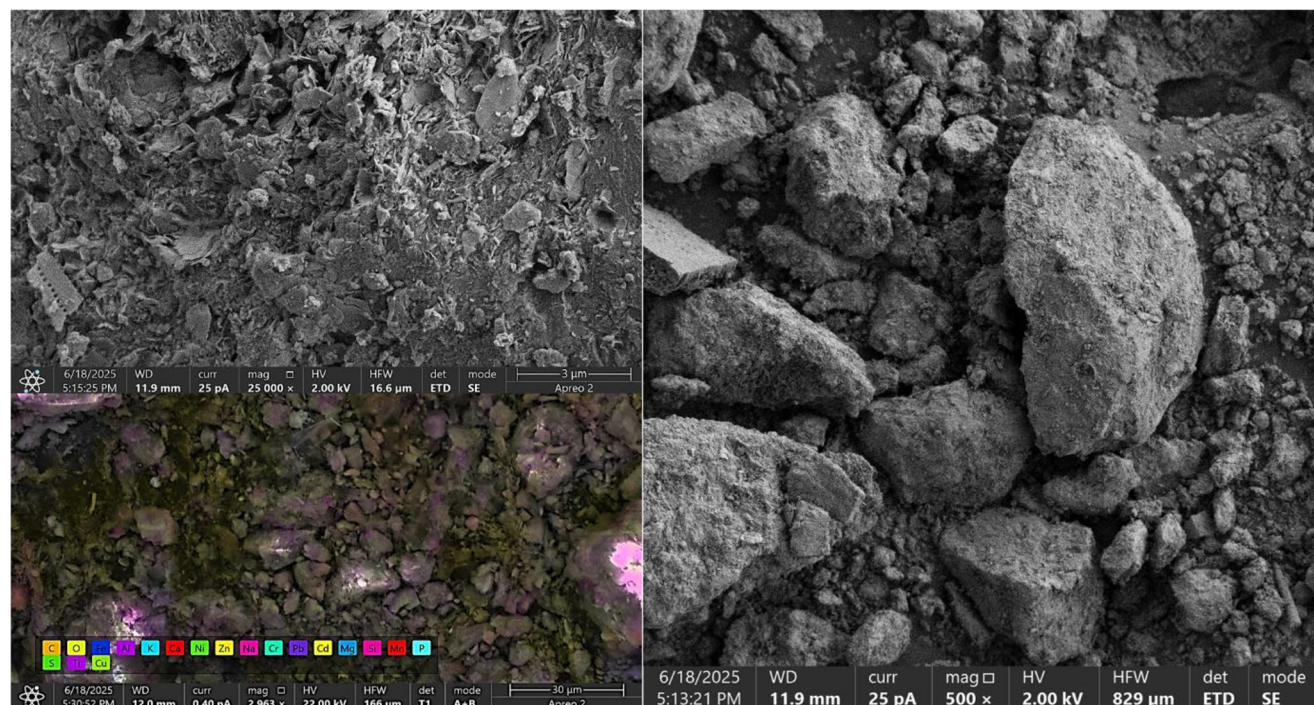


Fig. 3 SEM/EDS microphotographs of the PCWTS with energy-dispersive X-ray quantitative mapping

3.2 Analysis of water leachate obtained from the PCWTS

The characteristics of the PCWTS samples with a liquid-to-solid ratio of L: S=10:1 (L/kg dry matter) are presented in Table 5. The measured values were compared, as a reference, with the maximum permissible concentrations established for industrial wastewater, according to Polish regulations (Minister of Maritime Economy and Inland Navigation 2019).

Analysis of the leachate from the PCWTS sample revealed a neutral pH value of 7.2, which limits the mobility of heavy metals. The measured EC (154 $\mu\text{S}/\text{cm}$) and TDS (77 mg/L) were low, indicating the low salinity of the sample. The obtained values of COD and DOC were 112 mg O_2/L and 39 mg/L, respectively, suggesting a moderate

concentration of biodegradable organic matter. The concentrations of Cl^- (20 mg/L) SO_4^{2-} (42 mg/L) and F^- (0.2 mg/L) ions were below the threshold limit according to Polish regulations (Minister of Maritime Economy and Inland Navigation 2019).

Analysis of bioavailable nutrients revealed that the amounts of total nitrogen and total phosphorus were 5.3 and 0.33 mg/L, respectively. In contrast, the measured concentration of potassium in the water extract was 6.0 mg/L. Ca^{2+} and Mg^{2+} were detected at concentrations of 24 mg/L and 5 mg/L, respectively. The sludge sample had low contents of Al and Fe ions at levels below the permissible threshold (3 and 10 mg/L, respectively).

Analysis of leachate at a liquid-to-soil ratio of 10:1 L/kg obtained from PCWTS revealed that the concentrations of metals (Cd, Cr, Ni, Pb and Zn), which are crucial for soil

Table 5 Physiochemical parameters of the water leachate for the PCWTS

Parameter	Unit	Result	Limit values
pH	n.a.	7.20±0.05	6.5-9.0
Electrical conductivity (EC)	μS/cm	154 ± 8	n.a.
Total dissolved solids (TDS)	mg/L NaCl	77±4	35
Chemical oxygen demand (COD)	mgO ₂ /L	112±17	125
Dissolved organic carbon (DOC)	mg/L	39±5.8	30
Chloride (Cl ⁻)	mg/L	20±2	1000
Sulfate(VI) (SO ₄ ²⁻)	mg/L	42±6.3	500
Fluoride (F ⁻)	mg/L	0.20±0.03	25
Total N	mg/L	5.3±0.3	30
Total P	mg/L	0.33±0.02	2
Ca	mg/L	24.0±2.4	n.a.
Mg	mg/L	5.0±0.5	n.a.
K	mg/L	6.0 ± 1.2	80
Al	mg/L	<0.01	3
Fe	mg/L	<0.06	10
Cd	mg/L	<0.01	0.2
Cu	mg/L	0.07±0.02	0.5
Cr	mg/L	<0.05	0.1
Ni	mg/L	<0.05	0.5
Pb	mg/L	<0.07	0.5
Zn	mg/L	<0.01	2

n.a. not applicable

contamination, were below the detection limits, while the content of Cu (0.07 mg/L) was below the permissible concentration. This test is valuable and crucial for determining the bioavailability of toxic metals in the soil environment. On the basis of the criteria for the acceptance of wastewater discharged into the environment, the amounts of TDS and DOC exceed the limit values. Although the rest of the measured parameters meet the criteria, they do not fully reflect the long-term behavior of the PCWTS in land rehabilitation scenarios.

3.3 Risk assessment

3.3.1 Assessment of the potential ecological risk index (PERI)

The potential ecological risk of the HMs present in the PCWTS is presented in Fig. 4. The results of the PERI for green areas located in Poland revealed considerable ecological risk for the PCWTS. The primary cause of sludge pollution is Cd, which may pose a high risk of pollution

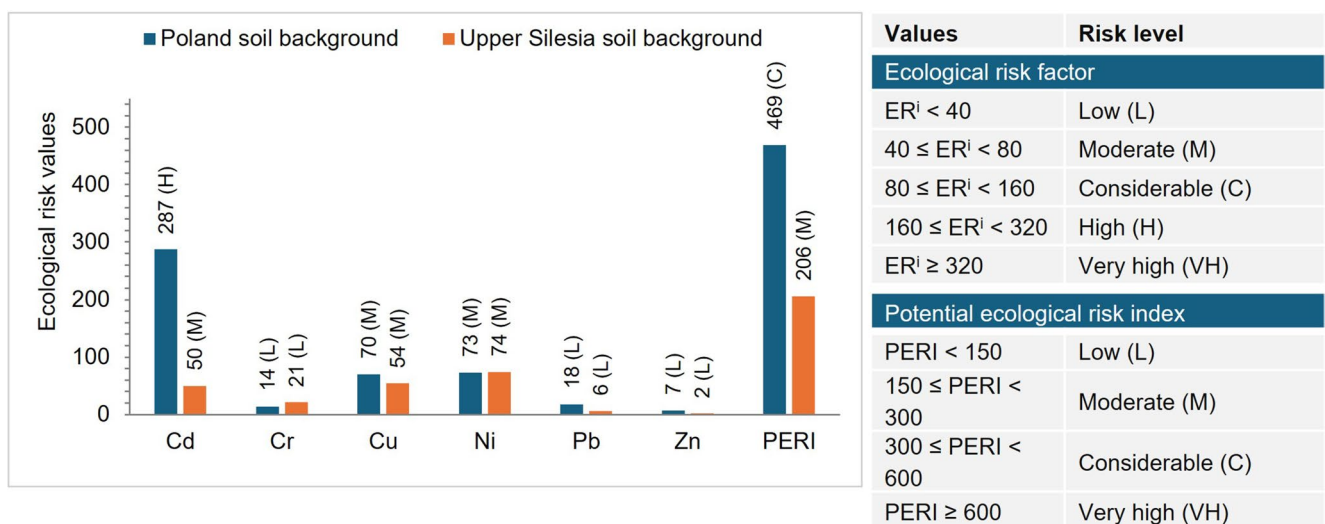


Fig. 4 Results of the ecological risk assessment for PCWTS, which is based on two soil backgrounds, associated with the ecological risk factor (ERⁱ) for selected metals (Cd, Cr, Cu, Ni, Pb and Zn) and the potential ecological risk index (PERI)

after the application of sludge to uncontaminated soils. The calculated ER_i values indicate that Cu and Ni are associated with moderate risk ($ER_i < 80$), whereas the pollution levels determined for Cr, Pb, and Zn are low ($ER_i < 40$). The results obtained from soils in the Upper Silesia region indicated a moderate potential ecological risk index for the PCWTS (PERI=206). The calculated values of the ecological risk factors were low ($ER_i < 40$) and moderate ($40 \leq ER_i < 80$) and varied in the order of $Ni > Cu > Cd > Cr > Pb > Zn$.

3.3.2 Assessment of contamination level

I_{geo} was used to assess the degree of PCWTS contamination by HMs. The results are presented in Fig. 5. The contamination levels for PCWTS according to the Polish soil background for grazing lands range from moderately polluted for Pb ($I_{geo} > 1$) to strongly polluted ($I_{geo} > 3$) for Cu and Ni. The calculated values of I_{geo} for Cd, Cr and Zn ranged between 2.23 and 2.65 and were assigned to intermediate class 3, i.e., moderately to strongly polluted. On the basis of the contamination levels and background of the soils in the Upper Silesia region, an unpolluted contamination class ($I_{geo} < 0$) was identified for Pb. Intermediate class 1, unpolluted to moderately unpolluted, was calculated for Cd and Zn ($I_{geo} > 1$). The highest level of pollution was observed for Ni ($I_{geo} > 3$), while the concentrations of Cr and Cu indicated moderate pollution ($I_{geo} > 2$).

3.4 Ecotoxicological risk assessment

The results of the ecotoxicological risk assessment, using test organisms representing different trophic levels are presented in Table 6. The results of the Phytotoxkit® tests for

the PCWTS water extract indicated that the inhibition values for *Lepidium sativum* and *Sinapis alba* were below the toxicity limit ($I < 30\%$). The same assessment of the non-toxic (30%) immobilization value was achieved for *Daphnia magna*. The observation of the luminescent bacteria revealed that the inhibitory effect on *Aliivibrio fischeri* was within the nontoxic range ($I < 20\%$) after 15 min. However, the same value after 30 min slightly exceeded 20% and was assigned to moderate toxicity. This suggests a time-dependent response, although the observed effect remains relatively low and does not indicate significant acute ecotoxicity under the tested conditions.

The results of microbiotests with higher-order plants on PCWTS solid samples with standardized control soil revealed negative growth inhibition. The measured root length for *Lepidium sativum* in the tested samples was lower by approximately 30%, and in the test with *Sinapis alba* seeds, it was 11% lower than that in the control soil. This means that both tests involved a nontoxic assessment of PCWTS as an amendment to soil for land rehabilitation.

A photographic illustration of the results of the micro-biotest Phytotoxkit® tests for the PCWTS water extract is presented in Fig. 6.

3.5 Pot test analysis

The experiments were conducted using three groups of soil samples, A1, A2 and A3, with different concentrations of PCWTS. The results of the pot test revealed that the average biomass weight increased with increasing amounts of PCWTS from 10 to 20 wt%. To illustrate the observed relationship between the PCWTS concentration in soils and the measured values of plant growth (Fig. 7), a quadratic trend

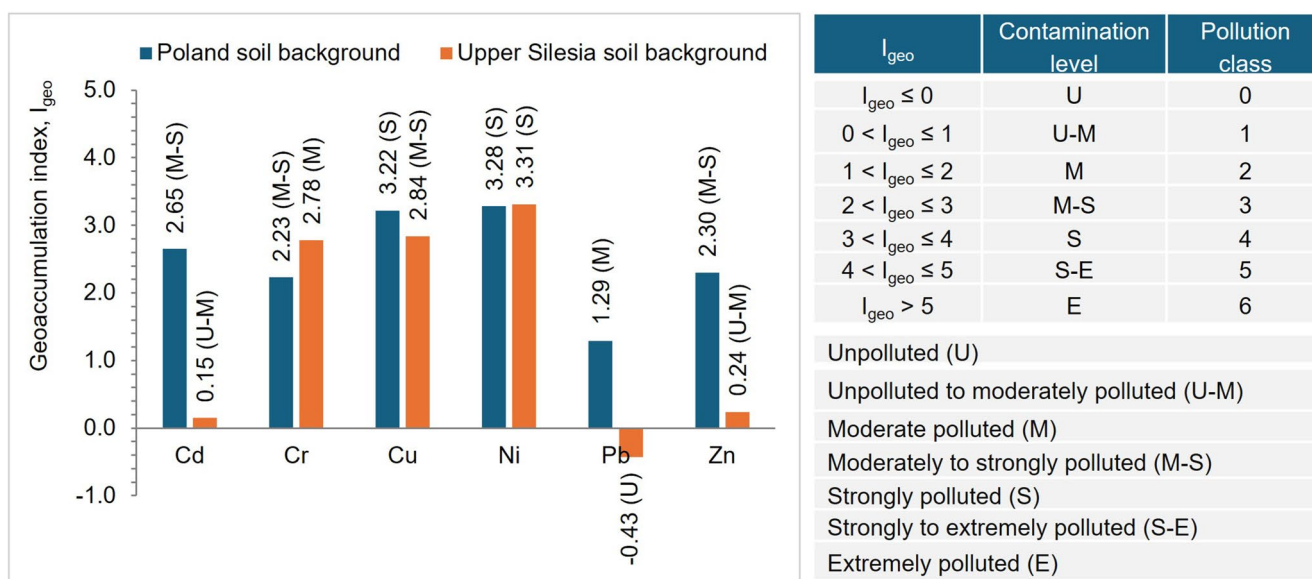


Fig. 5 Results of the pollution class for the PCWTS, based on the two soil backgrounds, associated with the geoaccumulation index (I_{geo})

Table 6 Results of the ecotoxicological tests for the PCWTS water extract

Test	Test organisms	Results (%)	Assessment (%)
Phytotoxkit® for liquid samples	<i>Lepidium sativum</i>	$I = 29.74 \pm 0.92$	$I \leq 30$, nontoxic
	<i>Sinapis alba</i>	$I = 27.25 \pm 0.99$	$30 \leq I < 50$, toxic $50 \leq I$, strongly toxic
Luminescent bacteria test kit LCK 480	<i>Aliivibrio fischeri</i>	$I_{15} = 19.33 \pm 0.20$	$I < 20$, nontoxic
		$I_{30} = 21.08 \pm 0.78$	$20 \leq I < 30$, moderately toxic $30 \leq I_m$, very toxic
Daphtoxkit F®	<i>Daphnia magna</i> S.	$I_m = 30 \pm 2.89$	$I_m < 50$, nontoxic
Phytotoxkit® for liquid samples	<i>Lepidium sativum</i>	$I = 29.91 \pm 1.47$	$I \leq 30$, nontoxic
	<i>Sinapis alba</i>	$I = 11.42 \pm 1.43$	$30 \leq I < 50$, toxic $50 \leq I$, strongly toxic

I Inhibition, I_m Immobilization

line was used to determine a high R^2 coefficient (0.96 for A1 and 0.99 for A2 and A3). However, owing to the limited number of amendments, the fitted curve should not be interpreted as evidence of an optimal PCWTS dose. ANOVA revealed significant differences between the soil samples with varying amounts of PCWTS for soils A1 and A2. The A1 sample without PCWTS application contained 0.647 g of biomass. The amendments of 10, 15, and 20 wt% PCWTS to soil A1 resulted in increases in plant growth from 0.757 to 0.987 to 2.889 g, respectively ($p < 0.001$, $F = 126.37$, $R^2 = 0.98$, $R^2_{adj} = 0.97$). A similar tendency for the growth of the grass mixture was observed for soil A2. The average weight of biomass measured for sample A2 with no application of PCWTS was zero and increased from 0.520 to 0.985 g after amendment with 10 to 20 wt% PCWTS, respectively ($p < 0.001$, $F = 28.64$, $R^2 = 0.91$, $R^2_{adj} = 0.88$). Compared with those of soils A1 and A2, the results of the pot test on soil A3 enriched with PCWTS were not significantly different ($p = 0.591$, $F = 0.675$, $R^2 = 0.20$). A low F statistic (0.675) indicates that the variability between groups is less than that within groups. Analysis between the groups of soil samples A3 revealed a slight increase in biomass weight, from 2.588 g with no amendment of PCWTS

to 2.858 and 3.031 g with the application of 10 and 20 wt% PCWTS, respectively. However, the observed differences between the values may be due to random variability.

4 Discussion

4.1 Advantages and limitations of PCWTS for land reclamation

The analysis revealed that the tested PCWTSs contained essential nutrients for plant growth and development, including N (9.7 mg/kg), P (2.3 mg/kg) and K (22.9 mg/kg). Although these total content values do not reflect their bioavailable fractions, the results presented in this paper are consistent with those obtained by other authors. A literature review revealed that the typical chemical macronutrients in aluminium sludge range from 1.06 to 8.68 g/kg for N, < 0.01–20.06 g/kg for K, and 0.89–3.9 g/kg for P (Pająk 2023). According to Ciula et al. (2024), the concentrations of the main macronutrients in postcoagulation sludge with the same coagulant (polyaluminium chloride) were 0.26 mg/kg for P_2O_5 and 2.36 mg/kg for K_2O . The



Fig. 6 The results of inhibition of plant growth according to Phytotoxkit® tests for PCWTS leachates (L: S=10:1) with the test organisms (a) *Lepidium sativum* (b) *Sinapis alba* L

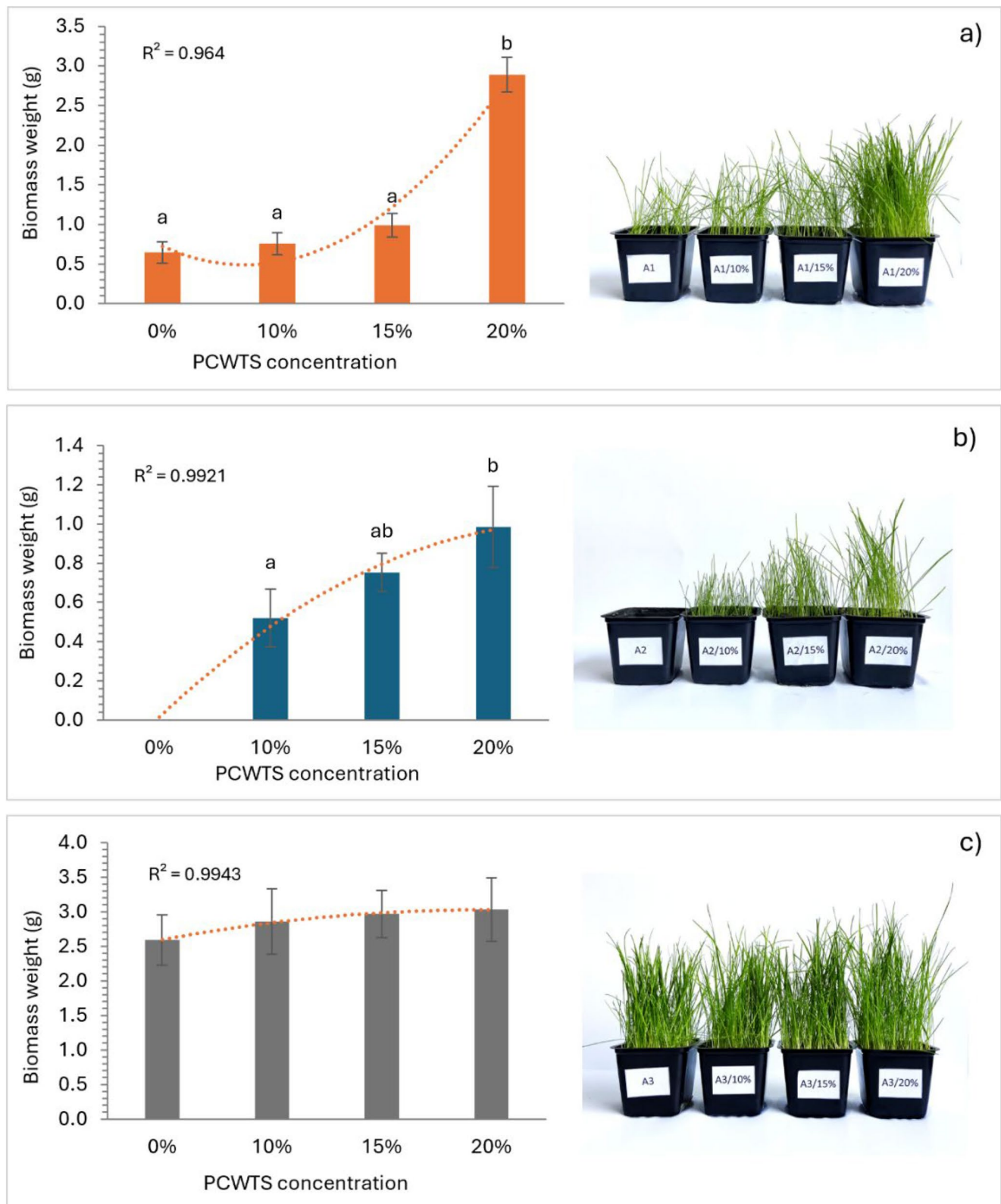


Fig. 7 Effects of growing a mixture of grass after 14 days on anthropogenic soils with amendments of 10, 15 and 20% PCWTS: (a) soil sample A1, (b) soil sample A2 and (c) soil sample A3. The same letters

(a, b) indicate that the groups are not significantly different from each other according to Tukey's HDS test ($p < 0.05$)

same authors reported that the nitrogen content determined via the Kjeldahl method was 7.5 g/kg, which is close to our results (9.7 g/kg). The total P concentration (2.3 g/kg) in the PCWTS was greater than that in samples A1–A3 (0.3–0.4 g/kg), which is much lower than that in typical sewage sludge from wastewater treatment plants (17.5–24.8 mg/kg) (Sichler et al. 2022; Kominko et al. 2017; Pesonen et al. 2016). Phosphorous is essential for plant growth, but its relatively low content in soil is beneficial for limiting the occurrence of eutrophication in water bodies (Muisa et al. 2020).

One of the major components in the residual from the coagulation process in the WTP is aluminium. Prehydrolysed forms of coagulants such as polyaluminium chloride generate lower concentrations of residual aluminium in treated water than nonhydrolysed salts do. The content of Al_2O_3 in the polyaluminium chloride coagulant was 22.6%, and its measured pH was 4.2 (Łukasiewicz and Rząsa 2017). Our study indicated that the Al content in PCWTS expressed in Al_2O_3 was 10.7%, which was relatively lower than that in typical sludge characterized in the literature. In comparison, the Al_2O_3 content in the most widely used alum sludge ($\text{Al}_2(\text{SO}_4)_3 \cdot 18\text{H}_2\text{O}$) ranges between 12.1 and 51.1% (Agyin-Birikorang and O'Connor 2009; Hovsepian and Bonzongo 2009; Pająk 2023).

Data from the literature confirm that prehydrolysed coagulants are considered more effective in the water treatment process and require lower dosages than nonhydrolysed coagulants do (Kłos and Gumińska 2010). The lower Al concentration in sludge is beneficial for its use in land reclamation and makes PCWTS a suitable amendment for soil. In accordance with Polish regulations concerning ground surface pollution (Minister of Environment 2016), Al is not listed as a toxic element in soil. However, some coagulants have shown that an increase in aluminium compounds in soil solutions is harmful to trees and agricultural plants (Kluczka et al. 2017). The toxic effects of Al on plants are associated with Al^{3+} cations, whose concentration increases when the soil pH is less than 5.5 (Zhao et al. 2018). In acidic soils, the concentration of Al^{3+} is greater than that in slightly alkaline soils. Insoluble aluminium compounds such as aluminium oxide (i.e., Al_2O_3) are considerably less toxic than soluble forms such as aluminium chloride (AlCl_3), aluminium nitrate ($\text{Al}(\text{NO}_3)_3$), aluminium acetate ($(\text{CH}_3\text{COO})_3\text{Al}$), and aluminium sulfate ($\text{Al}_2(\text{SO}_4)_3$) (US EPA 2003). Furthermore, some studies have reported that using sludge containing aluminium compounds as a soil ameliorant has no adverse effect on plant growth (Dasanayake et al. 2015). Kim et al. (2002) confirmed that the application of alum sludge to acidic soil with a pH of 4 in a greenhouse increased the buffer capacity, hydraulic conductivity, water-holding capacity, and phosphate adsorption

of the soil, leading to increases in biomass and root elongation. However, other studies reported different findings. For example, Dayton and Basta (2001) reported that the use of more air-dried sludge with $\text{Al}_2(\text{SO}_4)_3$ in farmlands may decrease crop yield. A laboratory study conducted by Agyin-Birikorang and O'Connor (2009) has confirmed that land application of freshly generated aluminium sludge has no harmful effect on plants or grazing animals but should be avoided because of the low availability of Al^{3+} . Nevertheless, in some cases, the high Al content in sludge may also cause adverse effects. The land application of aluminium sludge can affect the adsorption of phosphates and other plant nutrients in soil (Ahmad et al. 2016). The surface water quality criterion (Minister of Maritime Economy and Inland Navigation 2019) indicates that aluminium concentrations below 3 mg/L are generally considered not harmful to the aquatic environment or soil. However, at low pH values, aluminium occurs in the form of bioavailable and soluble Al^{3+} ions or hydroxocomplexes, posing a potentially lethal threat to aquatic organisms (Mortula et al. 2009). This is important in the context of soil remediation and waste management. The present research has revealed that the concentration of Al in the water extracts from PCWTS was 0.01 mg/L, indicating that there was no environmental risk to surface water or soil after disposal. In addition, prehydrolysed forms of coagulants, such as polyaluminium chloride, generate lower concentrations of residual aluminium in treated water than nonhydrolysed salts do, and the use of prehydrolysed coagulants results in a better coagulation process at a lower dosage (Kłos and Gumińska 2010).

Several researchers have demonstrated a significant correlation between soil pH and the phytotoxicity of Al species. The concentrations and toxicity of Al species in the soil environment depend on the pH level and chemical composition of the soil solution. Several soluble Al species, such as Al^{3+} , $\text{Al}(\text{OH})^{2+}$, $\text{Al}(\text{OH})^{2+}$, and $\text{Al}(\text{OH})^{4-}$, have been shown to occur when the soil pH is less than 5. Al dissolution occurs as a result of soil acidification, which leads to the release of phytotoxic Al^{3+} . At a pH < 4.3, trivalent Al^{3+} , the most abundant and toxic species, has the greatest impact on plant growth (Rahman et al. 2018; Kisiñerienė and Lapeikaitė 2015; Singh et al. 2017).

Although aluminium mobilization under strongly acidic conditions may represent a potential environmental risk, our results show (Table 2) that the Al concentration in the PCWTS (56.4 g/kg) is lower than that observed in two of the degraded soils tested in this study, i.e., A2 (84.7 g/kg) and A3 (85.2 g/kg). The concentration of Al in PCWTS is also lower than that in other sludge produced in other water treatment plants, which was 93.3 g/kg (Takao et al. 2024). Particular attention should be given to soil A2, which originates from a mining waste heap and is characterized

by strongly acidic conditions (pH=2.8). In this case, adding PCWTS is unlikely to increase the environmental risk of Al in the degraded area. In contrast, compared with the PCWTS, soil A1 (pH=5.2) is characterized by a lower Al concentration (15.4 g/kg). Therefore, under these conditions, the application of PCWTS should be considered more carefully. Owing to the variable composition of PCWTS resulting from many factors, its use for the reclamation of anthropogenically degraded soils must be preceded by a detailed analysis.

4.2 Ecotoxicological responses of PCWTS leachate in the context of literature data

In the literature, studies addressing the ecotoxicological assessment of postcoagulation sludge derived from the water treatment process are lacking. However, several studies have investigated the ecotoxicity of sewage sludge or leachates from municipal waste, and microbiological tests have been used to assess their potential effects on soil or aquatic microorganisms (Table 7). Sotero-Santos et al. (2005) analysed the toxicity of alum sludge to *Daphnia similis*, and no toxic effect on the mortality of the test organism was observed. Yuan et al. (2020) reported that dewatered water treatment residue has no toxic effect on the survival, growth, reproduction, body morphology, or antioxidant enzyme activity of *Daphnia magna*. These findings are consistent with the results obtained in the present study. An acute immobilization test of PCWTS, which uses *Daphnia magna*, indicated 30% toxicity after 24 and 48 h of exposure, which corresponds to moderate toxicity, according to the Guidelines for the Testing of Chemicals (OCDE 2004).

Moreover, the results of the remaining toxicity tests on the water extract from the PCWTS revealed in this study were also negative. These findings indicate that the percentage immobilization ($I_{15,30} < 50\%$) of *Aliivibrio fischeri* and the percentage inhibition of root growth of *Sinapis alba* and *Lepidium sativum* ($I < 50\%$) were less toxic. In comparison with a study carried out by other authors, at the same ratio of 1:10 kg/L, the growth inhibition of test plants in a sample of sludge from pulp waste stimulated the growth of *L. sativum* (86.63%) and inhibited the growth of *S. alba* (62.92%) (Novak 2023). The toxicity of two types of sewage sludge from a wastewater treatment plant (dewatered and anaerobically stabilized) in relation to *Sinapis alba* was studied by Adamcová et al. (2016a, 2016b). The results revealed that both sludges at a concentration of 10% were toxic, and the average growth inhibition factors were 96.9% and 77.1%, respectively. Aquatic toxicity tests were carried out by Kočí et al. (2010) using eluates (1:10 kg/L) from water-treated sludge. The calculated value of inhibition for *Sinapis alba*

root elongation was 18%, which is comparable to our results (27.2%). Compared with wastewater treatment sludge, the data in Table 9 show adverse effects (more than 50% inhibition) on the growth of *Sinapis alba* and *Lepidium sativum* in sewage sludge (Adamcová et al. 2016a) or industrial wastewater (Novak 2023). One of the reasons for the toxic effects on plant growth and development is the salinity of the soil, expressed as the electrical conductivity of its water leachate. According to the general classification (Miller and Donahue 1995), slight soil salinity may be observed when the EC exceeds 2 mS/cm, and high salinity may occur when the EC is greater than 8 mS/cm. Under these conditions, sprouting and biomass growth may be restricted. In general, the main ions associated with soil salinity are K^+ , Na^+ , SO_4^{2-} and Cl^- . Our research confirmed that the measured concentration of dissolved ions in water leachate from PCWTS is significantly low (77 mg NaCl/L for TDS, 20 mg/L for SO_4^{2-} and 42 mg/L for Cl^-), and the measured value of EC is 0.154 mS/cm.

The results of the pot test in which a mixture of grass was added to anthropogenic soils suggest that there was no acute phytotoxic effect at PCWTS amendment rates ranging from 10 to 20%. A positive impact on biomass growth was observed across all the tested soil samples (A1–A3) collected from areas degraded by industrial activity. These findings indicate that PCWTS may improve the quality of contaminated soils. However, as the pot test was conducted under short-term conditions only, further investigations are still required. To fully evaluate the environmental safety and suitability of PCWTS for soil application, long-term studies that track the accumulation of potentially toxic metals in plant tissues and changes in their concentrations in the soil over time would be beneficial.

4.3 Ecological risk of PCWTS and regulatory thresholds of heavy metals

Another reason is the high concentration of toxic substances, including HMs. Water-treated sludge contains substances that may originate from both the raw water and the reagents used in the treatment process (Piedgoń 2024; Fiore et al. 2022). Studies indicate that the majority of postcoagulation sludge does not contain excessive amounts of toxic substances that would classify them as hazardous. However, when sludge contains high concentrations of heavy metals, it may pose a threat not only to soils but also to groundwater and even surface water.

Specific regulatory thresholds for applying water treatment sludge to soils are not available. Therefore, the determined HM concentrations in the tested PCWTS were compared with the maximum contents of toxic elements in agricultural or industrial soils (Minister of Environment

2016) and the regulatory limits established for sewage sludge applied for land reclamation (Minister of Environment 2015). The presence of heavy metals in PCWTS is significantly influenced by the quality of the water used for treatment and by the type of coagulant used. Tests of post-coagulation sludge conducted by other authors (Ciula et al. 2024; Alvarenga et al. 2018; Ippolito et al. 2011; Takao et al. 2024) revealed that the concentrations of selected HMs, described as hazardous for soil (Minister of Environment, 2016), ranged from 20.4 to 624 mg/kg for Cu, 6.7 to 38.5 mg/kg for Cr, 0.12 to 1.8 mg/kg for Cd, 18.2 to 43.58 mg/kg for Ni, 2.7 to 22 mg/kg for Pb, and 28.9 to 98 mg/kg for Zn. The concentrations of the same HMs in this study were relatively high. However, the total concentration of HMs in the tested PCWTS was below the permissible levels in terms of sewage sludge application for land reclamation in nonagricultural areas. In accordance with the Polish requirements (Minister of Environment 2015) and the European Council Directive 86/278/EEC (European Commission 1986), which establish the criteria for the use of municipal sewage sludge which may be applied to land, the limit values of toxic elements are as follows (in mg/kg): 25 for Cd, 1200 for Cu, 1000 for Cr, 400 for Ni, 1000 for Pb, and 3500 for Zn. An ecological risk assessment revealed that the use of PCWTS as a soil-improving agent may not pose a high risk of HMs contamination in areas degraded by industrial activity as evaluated against the background concentrations of soils in the Upper Silesia region. The calculated risk level for green areas, excluding agricultural soils, located in Poland was considerable, with a total PERI value of less than 600. However, considering the background values for soil in the Upper Silesia region, the risk was moderate, and the PERI value did not exceed 300. A comprehensive review of subject literature revealed a lack of data concerning ecological risk assessments for water treatment sludge. Ashong et al. (2024) demonstrated that the presence of Mn, Pb and Zn in post-treatment sludge from the WTP was very low and did not have a toxic effect on either the environment or human health. For comparison purposes, Table 8 presents the values of the ecological risk assessment indicators for sludge from water and wastewater treatment. The highest risk level, with a PERI greater than 1000, was indicated for sewage sludge. The analysis revealed that cadmium (Cd) was the most harmful element. These results are consistent with our analysis of green areas in Poland, where the calculated value of ER^i for Cd was significant (287). However, the aim of the present research was to assess the feasibility of using PCWTS as a material for land reclamation in anthropogenically degraded soil in the case of which the Cd concentration may exceed pre-existing levels before the addition of PCWTS. Compared with the sewage sludge sample, the PCWTS sample was characterized by a higher content of Ni

(74 mg/kg), but its ER^i was moderate. Although the SEM/EDS results provided complementary information on the elemental composition of the PCWTS sample surface, this method does not allow the detection of potentially toxic elements at trace concentrations, which may pose an environmental risk. However, the high contents of aluminium, silicon, and iron observed seem to suggest that the PCWTS sample possesses considerable mineral purity and consists mainly of inorganic residue. Our results align with those of Saleman et al. (2023), who reported that the weight% of sludge from a water treatment plant ranged mainly from 7.25 to 26.85% carbon, 29.04–40.93% oxygen, 6.83–7.59% aluminium, and 7.33–8.70% silicon.

To assess the mobility of toxic metals in soil, additional studies are still required. The use of sequential metal extraction procedures (Rauret et al. 2001) in sludge and its mixtures with anthropogenic soils may constitute a crucial aspect in this case. Such analyses could provide deeper insight into the specifications of metals and their potential mobility in response to long-term environmental processes, including weathering, changes in pH, and redox conditions. An additional limitation of this study consists in the fact that the leaching tests were carried out separately for samples of degraded soils (A1–A3) and for PCWTS, whereas no analysis was conducted for soil with a 10–20% PCWTS amendment. Therefore, assessing the mobility and bioavailability of HMs and Al in soil mixtures, particularly under strongly acidic conditions, requires additional research.

5 Conclusion

PCWTS is a semisolid byproduct generated during the treatment of surface water to obtain high-quality drinking water. It is a mixture of water, organic matter, inorganic particles, and treatment chemicals such as coagulants and flocculants. PCWTS mainly results from coagulation, flocculation, and sedimentation processes and typically contains aluminium or iron hydroxides, depending on the applied coagulant. HMs and micropollutants may limit reuse, and improper disposal may contaminate soil, as well as ground and surface water. For this reason, in many countries, regulations often dictate disposal and reuse strategies. The composition of PCWTS together with its physicochemical and ecotoxicological properties are of key importance in the process of assessing its environmental impact and delimitating its practical use. However, the results of this short-term study indicate the potential suitability of PCWTS as an amendment for land reclamation in low-quality and industrially degraded soils. Analysis of the PCWTS, water extraction, and risk assessment (ER^i , PERI, and I_{geo}) did not reveal a high level of risk associated with the application of sediments to soils under

Table 8 Concentrations of heavy metals and values of ecological risk indicators (ERⁱ, PERI) for sediments from various sources

Tested material	HMs, mg/kg	Ecological risk assessment ^(a)		Resources
		ER ⁱ	PERI	
PCWTS	Cd 2.2 Cr 51 Cu 75 Ni 74 Pb 49 Zn 184	Cd 287 (H) and 50 (M) Cr 14 and 21 (L) Cu 70 and 54 (M) Ni 73 and 74 (M) Pb 18 and 6 (L) Zn 7 and 2 (L)	469 (C) and 206 (M)	This study
Posttreatment sludge from the water treatment plant, Ghana	Mn 7.8 Pb 0.8 Zn 0.3	Mn 0.05 (L) Pb 0.49 (L) Zn 0.04 (L)	0.58 (L)	(Ashong et al. 2024)
Sewage sludge from treatment plant, Poland	Cd 2.4 Cr 142.6 Cu 196 Ni 43.9 Pb 56.8 Zn 2777	Cd 800 (VH) Cr 86.4 (M) Cu 326.7 (VH) Ni 91.5 (M) Pb 26.3 (L) Zn 143.9 (H)	1475 (VH)	(Kowalik et al. 2019)
Sediment from urban river, China	Cu 45.3 Cr 35.6 Cd 2.9 Ni 35.1 Pb 21.9 Zn 205.7	Cu 540 (VH) Cr 1 (L) Cd 9 (L) Ni 7 (L) Pb 5 (L) Zn 3 (L)	565 (C)	(Liu et al. 2021)
Sewage sludge from municipal wastewater treatment plants, Poland	Cd 1.4–4.6 Cr 17.8–66.1 Cu 79.7–174.9 Ni 17.0–64.8 Pb 39.6–170.9 Zn 774.2–2248	Cd 435 (VH)–1385 (VH) Cr 0.4 (L)–1.3 (L) Cu 7.2 (L)–15.9 (L) Ni 3.6 (L)–16.2 (L) Pb 13.2 (L)–57 (M) Zn 11.1 (L)–32.1 (L)	576 (C)–1582 (VH)	(Tyrtla 2020)
Sewage sludge from municipal wastewater treatment plants, China	As 11.6 Cu 100.5 Cr 147.7 Ni 34.3 Pb 11.6 Zn 331.5	As 12.7 (L) Cu 21.9 (L) Cr 5.3 (L) Ni 5.7 (L) Pb 3.9 (L) Zn 5.2 (L)	245 (M)	(Duan et al. 2017)

^(a)–Risk levels: (L)–low, (M)–moderate, (C)–considerable, (H)–high, (VH)–very high

the tested conditions. Nevertheless, although the leachate met the relevant criteria, providing general recommendations requires long-term field data. Similar conclusions can be drawn from the results of ecotoxicological studies using selected test organisms (*Aliivibrio fischeri*, *Daphnia magna*, *Lepidium sativum*, and *Sinapis alba*). The results revealed the nontoxicity of PCWTS and its water extracts in the case of all the microbioassays. However, the PCWTS results from other water treatment plants should be further investigated. One of the reasons is that the HMs content in PCWTS strongly depends on the source water, the nature of the catchment area and the coagulant used. In turn, the results of the short-term pot tests using a grass mixture demonstrate that the effect of PCWTS amendment on increasing biomass is proportional to the dose used. Hence, despite the presence of minor heavy metals, PCWTS can be safely reused in soil, resulting in nontoxicity and beneficial effects on plant growth and soil quality. Therefore, in terms of the presence of heavy metals, the application of sludge is more justified at anthropogenic sites in Upper Silesia than in Polish soils in general. One limitation of this study is the lack of long-term field experiments, along with assessing the accumulation of toxic metals in plant tissues and recultivated soils. To confirm the use of PCWTS for land reclamation at anthropogenic sites characterized by strongly acidic properties, such as mining waste heaps, further research under low-pH conditions is crucial.

The use of PCWTS aligns harmoniously with the concepts of sustainable development and circular economy. However, pollution control and the potential development of technologies that can safely return these materials to the environment as recultivated soil are of vital importance. This work can constitute an important step in the discussion, design, planning, testing, and evaluation of the beneficial use of PCWTS support for more sustainable management practices and the implementation of circular economy principles. These findings highlight that the application of PCWTS in anthropogenic soils may contribute to sustainable land management, particularly in regions affected by long-term industrial degradation.

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References

- Adamcová D, Vavrková MD, Břoušková E (2016a) The toxicity of two types of sewage sludge from wastewater treatment plant for plants. *J Ecol Eng* 17:33–37. <https://doi.org/10.12911/2298993/62283>
- Adamcová D, Vavrková MD, Bartoň S, Havlíček Z, Břoušková E (2016b) Soil contamination in landfills: a case study of a landfill in Czech Republic. *Solid Earth* 7:239–247. <https://doi.org/10.5194/se-7-239-2016>
- Agyin-Birikorang S, O'Connor GA (2009) Aging effects on reactivity of an aluminum-based drinking-water treatment residual as a soil amendment. *Sci Total Environ* 407:826–834. <https://doi.org/10.1016/j.scitotenv.2008.09.025>
- Ahmad T, Ahmad K, Alam M (2016) Sustainable management of water treatment sludge through 3'R' concept. *J Clean Prod* 124:1–13. <https://doi.org/10.1016/j.jclepro.2016.02.073>
- Alvarenga P, Palma P, Gonçalves AP, Baião N, Fernandes RM, de Varennes A, Vallini G, Duarte E, Cunha-Queda AC (2008) Assessment of chemical, biochemical and ecotoxicological aspects in a mine soil amended with sludge of either urban or industrial origin. *Chemosphere* 72:1774–1781. <https://doi.org/10.1016/j.chemosphere.2008.04.042>
- Alvarenga P, Ferreira C, Mourinha C, Palma P, de Varennes A (2018) Chemical and ecotoxicological effects of the use of drinking-water

- treatment residuals for the remediation of soils degraded by mining activities. *Ecotoxicol Environ Saf* 161:281–289. <https://doi.org/10.1016/j.ecoenv.2018.05.094>
- Ashong GW, Ababio BA, Kwaansa-Ansah EE, Gyabeng E, Nti SO (2024) Human and ecotoxicological risk assessment of heavy metals in polymer post-treatment sludge from Barekese Drinking Water Treatment Plant, Kumasi. *Toxicol Rep* 12:404–413. <https://doi.org/10.1016/j.toxrep.2024.03.010>
- Ashworth AJ, Chastain JP, Moore PA (2020) Nutrient characteristics of poultry manure and litter. In: Waldrup HM, Pagliari PH, He Z (eds) *Animal manure: production, characteristics, environmental concerns, and management*. ASA and SSSA, Madison, WI, pp 63–87. <https://doi.org/10.2134/asapecpub67.c5>
- Balkaya M (2016) Evaluation of the use of alum sludge as hydraulic barrier layer and daily cover material in landfills: a finite element analysis study. *Desalin Water Treat* 57:2400–2412. <https://doi.org/10.1080/19443994.2015.1005154>
- Bartoszewski K (1996) *Waste water treatment and waste water recovery. Renewal of water. Theoretical basis of processes*. Wrocław University of Science and Technology, Wrocław
- Bérubé D, Dorea CC (2008) Optimizing alum coagulation for turbidity, organics, and residual Al reductions. *Water Sci Technol Water Supply* 8:505–511. <https://doi.org/10.2166/ws.2008.090>
- Boudjabi S, Chenchouni H (2021) On the sustainability of land applications of sewage sludge: How to apply the sewage biosolid in order to improve soil fertility and increase crop yield? *Chemosphere* 282:131122. <https://doi.org/10.1016/j.chemosphere.2021.131122>
- Chen L, Wang G, Wu S, Xia Z, Cui Z, Wang C, Zhou S (2019) Heavy metals in agricultural soils of the Lihe River Watershed, East China: spatial distribution, ecological risk, and pollution source. *Int J Environ Res Public Health* 16:2094. <https://doi.org/10.3390/ijerph16122094>
- Ciuła J, Wiewiórska I, Kulczycka J, Willson J (2024) Analysis of the physical and chemical composition of sludge from the water treatment plant. *Annu Set Environ Prot* 26:479–492. <https://doi.org/10.54740/ros.2024.045>
- Dassanayake KB, Jayasinghe GY, Surapaneni A, Hetherington C (2015) A review on alum sludge reuse with special reference to agricultural applications and future challenges. *Waste Manag* 38:321–335. <https://doi.org/10.1016/j.wasman.2014.11.025>
- Dayton EA, Basta NT (2001) Characterization of drinking water treatment residuals for use as a soil substitute. *Water Environ Res* 73:52–57. <https://doi.org/10.2175/106143001x138688>
- Duan B, Zhang W, Zheng H, Wu C, Zhang Q, Bu Y (2017) Disposal situation of sewage sludge from municipal wastewater treatment plants (WWTPs) and assessment of the ecological risk of heavy metals for its land use in Shanxi, China. *Int J Environ Res Public Health* 14:823. <https://doi.org/10.3390/ijerph14070823>
- EMPPPO (2007) European and Mediterranean Plant Protection Organization. Phytotoxicity assessment. Efficacy evaluation of plant protection products. EPPO Bull 37:4–10. <https://doi.org/10.1111/j.1365-2338.2007.01067.x>
- European Commission (1986) Council Directive 86/278/EEC of June 1986 on the protection of the environment, and in particular of the soil, when sewage sludge is used in agriculture, Official Journal of the European Communities. <https://doi.org/2004R0726-v.7of05.06.2013>
- European Commission (2019) European Economic and Social Committee and the Committee of the Regions on the Implementation of the Circular Economy Action Plan. Report from the Commission to the European Parliament, the Council. COM/2019/190, 2019.
- Fiore FA, Rodgher S, Koga Ito CY, Bardini VSDS, Klinsky LMG (2022) Water sludge reuse as a geotechnical component in road construction: Experimental study. *Clean Eng Technol* 9:100512. <https://doi.org/10.1016/j.clet.2022.100512>
- Gong C, Wen L, Lu H, Wang S, Liu J, Xia X, Liao Z, Wangzha D, Zhaxi W, Tudan J, Tan C (2024) Ecological, environmental risks and sources of potentially toxic elements in soils of Yangtze river source region, Qinghai-Tibet Plateau. *Environ Geochem Health* 46:460. <https://doi.org/10.1007/s10653-024-02161-6>
- Gworek B, Baczewska-Dąbrowska AH, Kalinowski R, Górka EB, Rekosz-Burlaga H, Olejniczak I, Chmielewski J, Dmuchowski W (2024) Ecological risk assessment based on the TRIAD approach in an area contaminated by the metallurgical and mining industries. *J Elem* 29:99–121. <https://doi.org/10.5601/jelem.2023.28.4.2439>
- Hakanson L (1980) An ecological risk index for aquatic pollution control. A sedimentological approach. *Water Res* 14:975–1001. [https://doi.org/10.1016/0043-1354\(80\)90143-8](https://doi.org/10.1016/0043-1354(80)90143-8)
- Hovsepian A, Bonzongo JCJ (2009) Aluminum drinking water treatment residuals (Al-WTRs) as sorbent for mercury: implications for soil remediation. *J Hazard Mater* 164:73–80. <https://doi.org/10.1016/j.jhazmat.2008.07.121>
- Ippolito JA, Barbarick KA, Elliott HA (2011) Drinking Water Treatment Residuals: A Review of Recent Uses. *J Environ Qual* 40:1–12. <https://doi.org/10.2134/jeq2010.0242>
- Jiang JQ, Graham NJD (1998) Pre-polymerised inorganic coagulants and phosphorus removal by coagulation - A review. *Water SA* 24:237–244
- Kalka J (2012) Landfill leachate toxicity removal in combined treatment with municipal wastewater. *Scientific World Journal* 2012:202897. <https://doi.org/10.1100/2012/202897>
- Kim JG, Lee SS, Moon HS, Kang IM (2002) Land application of alum sludge from water purification plant to acid mineral soil treated with acidic water. *Soil Sci Plant Nutr* 48:15–22. <https://doi.org/10.1080/00380768.2002.10409166>
- Kisnierené V, Lapeikaitė I (2015) When chemistry meets biology: The case of aluminium—a review. *Chemija* 26:148–158. <https://doi.org/10.6001/chemija.2015.26.3.2>
- Kłos M, Gumińska J (2010) New approach to evaluation of effectiveness of in-bed coagulation. *Archit Civ Eng Environ* 1:103–108
- Kluczka J, Zołotajkin M, Ciba J, Staroń M (2017) Assessment of aluminum bioavailability in alum sludge for agricultural utilization. *Environ Monit Assess* 189:1–8. <https://doi.org/10.1007/s10661-017-6133-x>
- Kočí V, Mocová K, Kulovaná M, Vosáhllová S (2010) Phytotoxicity tests of solid wastes and contaminated soils in the Czech Republic. *Environ Sci Pollut Res* 17:611–623. <https://doi.org/10.1007/s11356-009-0214-5>
- Kominko H, Gorazda K, Wzorek Z, Wojtas K (2018) Sustainable management of sewage sludge for the production of organo-mineral fertilizers. *Waste Biomass Valorization* 9:1817–1826. <https://doi.org/10.1007/s12649-017-9942-9>
- Kowalik R, Gawdzik J, Gawdzik B (2019) Risk analysis of accumulation of heavy metals from sewage sludge in soil from the sewage treatment plant in Starachowice. *Struct Environ* 11:287–295. <https://doi.org/10.30540/sae-2019-022>
- Leszczyńska M (2009) The harmfulness of the water treatment process residuals. *Water Supply Water Qual*. In: Sozański MM (ed) *Water supply and water quality*, vol 2, *Polskie Zrzeszenie Inżynierów i Techników Sanitarnych*, Poznań, pp 249–260
- Lis J, Pasieczna A (1995) *Geochemical Atlas Atlas Up Silesia* 1:200000
- Liu D, Wang J, Yu H, Gao H, Xu W (2021) Evaluating ecological risks and tracking potential factors influencing heavy metals in sediments in an urban river. *Environ Sci Eur* 33. <https://doi.org/10.1186/s12302-021-00487-x>
- Lue-Hing C (2024) *Municipal sewage sludge management: a reference text on processing, utilization and disposal*. Volume 4. Second Edition, CRC Press, Boca Raton. <https://doi.org/10.1201/9781003579007>

- Łukasiewicz E (2016) Post-coagulation sludge management for water and wastewater treatment with focus on limiting its impact on the environment. *Econ Environ Stud* 16:831–841
- Łukasiewicz E, Rząsa M (2017) Examination of coagulant additives on qualitative composition of selected thermal waters. *E3S Web Conf.* 19:0–3. <https://doi.org/10.1051/e3sconf/20171902013>
- Mantis I, Voutsas D, Samara C (2005) Assessment of the environmental hazard from municipal and industrial wastewater treatment sludge by employing chemical and biological methods. *Ecotoxicol Environ Saf* 62:397–407. <https://doi.org/10.1016/j.ecoenv.2004.12.010>
- Matilainen A, Lindqvist N, Tuhkanen T (2005) Comparison of the efficiency of aluminium and ferric sulphate in the removal of natural organic matter during drinking water treatment process. *Environ Technol* 26:867–876. <https://doi.org/10.1080/09593332608618502>
- McCauley A, Jones C, Jacobsen J (2011) Plant nutrient functions and deficiency and toxicity Symptoms. *Nutr Manag Modul* 9:1–16
- Mensah-Akutteh H, Buamah R, Wiafe S, Nyarko KB (2022) Raw water quality variations and its effect on the water treatment processes. *Cogent Eng* 9:2122152. <https://doi.org/10.1080/23311916.2022.2122152>
- Miller R, Donahue R (1995) *Soils in our Environment: An Introduction to Soils and Plant Growth*, 7th Revised edn. Cambridge University Press, Prentice-Hall
- Minister of Environment (2016) Regulation of the Minister of Environment on the manner of conducting assessment of pollution of the ground surface. *J Laws Repub Pol Item* 1395
- Minister of Maritime Economy and Inland Navigation (2019) Regulation of 12 July 2019 on substances particularly harmful to the aquatic environment and conditions to be met when discharging wastewater into waters or into the ground, and when discharging rainwater or meltwater into waters or water facilities. *J Laws Repub Pol Item* 1311
- Minister of Environment (2015) Regulation of the Ministry of Environment concerning the sewage sludge. *J Laws Repub Pol Item* 257
- Mohamed OA, El-Gamal SMA, Farghali AA (2022) Utilization of alum sludge waste for production of eco-friendly blended cement. *J Mater Cycles Waste Manag* 24:949–970. <https://doi.org/10.1007/s10163-022-01369-x>
- Mortula M, Bard SM, Walsh ME, Gagnon GA (2009) Aluminum toxicity and ecological risk assessment of dried alum residual into surface water disposal. *Can J Civ Eng* 36:127–136. <https://doi.org/10.1139/S08-042>
- Muisa N, Nhapi I, Ruziwa W, Manyuchi MM (2020) Utilization of alum sludge as adsorbent for phosphorus removal in municipal wastewater: A review. *J Water Process Eng* 35:101187. <https://doi.org/10.1016/j.jwpe.2020.101187>
- Muller G (1969) Index of geoaccumulation in sediments of the Rhine River. *GeoJournal* 2:108–118
- Nguyen MD, Thomas M, Surapaneni A, Moon EM, Milne NA (2022) Beneficial reuse of water treatment sludge in the context of circular economy. *Environ Technol Innov* 28:102651. <https://doi.org/10.1016/j.eti.2022.102651>
- Novak O (2023) Phytotoxicity assessment of wastewater from industrial pulp production. *Eng Proc* 57:30. <https://doi.org/10.3390/enproc2023057030>
- Nowacka A, Włodarczyk-Makuła M, Macherzyński B (2014) Comparison of effectiveness of coagulation with aluminum sulfate and pre-hydrolyzed aluminum coagulants. *Desalin Water Treat* 52:3843–3851. <https://doi.org/10.1080/19443994.2014.888129>
- OCDE (2004) OECD 202 Guidelines for testing of chemicals. *Daphnia* sp. acute immobilisation test. 2004. Paris, France. 2006
- Ouyang Y, Nkedi-Kizza P, Wu QT, Shinde D, Huang CH (2006) Assessment of seasonal variations in surface water quality. *Water Res* 40:3800–10. <https://doi.org/10.1016/j.watres.2006.08.030>
- Pająk M (2023) Alum sludge as an adsorbent for inorganic and organic pollutants removal from aqueous solutions: a review. *Int J Environ Sci Technol* 20:10953–10972. <https://doi.org/10.1007/s13762-023-04854-4>
- Parliament of the Republic of Poland (2012) Announcement of the Parliament of the Republic of Poland of July 7, 2023, on the publication of the consolidated text of the Waste Act, *J. Laws Repub. Pol.*, 1587. Available online: <https://isap.sejm.gov.pl/isap.nsf/DocDetails.xsp?id=WDU20130000021> (accessed on February 13, 2026)
- Pasieczna A, Markowski W (2014) Geochemical mapping of agricultural soils and grazing land in Poland - National Report. Warsaw. Available online: (accessed on February 12, 2026). https://mapgeochem.pgi.gov.pl/wp-content/uploads/2021/06/gemas_tab_1-1.pdf
- Pesonen J, Kuokkanen V, Kuokkanen T, Illikainen M (2016) Co-granulation of bio-ash with sewage sludge and lime for fertilizer use. *J Environ Chem Eng* 4:4817–4821. <https://doi.org/10.1016/j.jec.2015.12.035>
- Piedgoń I (2024) Variability of drinking water quality on the basis of analysis of qualitative monitoring from a selected water supply network located in South-Eastern Poland. *Water* 16:3355. <https://doi.org/10.3390/w16233355>
- Płonka I, Pieczykolan B, Thomas M (2026) Electromagnetic disintegration of water treatment sludge: physicochemical changes and leachability assessment. *Appl Sci* 16:110. <https://doi.org/10.3390/app16010110>
- Rahman MA, Lee S-H, Ji HC, Kabir AH, Jones CS, Lee K-W (2018) Importance of mineral nutrition for mitigating aluminum toxicity in plants on acidic soils: current status and opportunities. *Int J Mol Sci* 19:3073. <https://doi.org/10.3390/ijms19103073>
- Rauret G, López-Sánchez JF, Lück D, Yli-Halla M, Muntau H, Quevauviller P (2001) The certification of the extractable contents (mass fractions) of Cd, Cr, Cu, Ni, Pb and Zn in freshwater sediment following a sequential extraction procedure BCR-701, Report EUR 19775 EN, European Commission, Brussels. <https://ec.europa.eu/jrc/en/publication/eur-scientific-and-technical-research-reports/additional-certification-pb-mass-fraction-bcr-320r-channel-sediment> (accessed on December 23, 2026)
- Saleman AR, Jumaidin R, Sultan AAM, Azlan UAA, Abdul Munir F, AbGhani A.F (2023) Chemical and microscopic analysis of water treatment sludge (WTS) from Bukit Sebukor water treatment plant. *Malaysian J Microsc* 19:33–41
- Sichler TC, Montag D, Barjenbruch M, Mauch T, Sommerfeld T, Ehm JH, Adam C (2022) Variation of the element composition of municipal sewage sludges in the context of new regulations on phosphorus recovery in Germany. *Environ Sci Eur* 34:84. <https://doi.org/10.1186/s12302-022-00658-4>
- Singh S, Tripathi DK, Singh S, Sharma S, Dubey NK, Chauhan DK, Vaculik M (2017) Toxicity of aluminium on various levels of plant cells and organism: A review. *Environ Exp Bot* 137:177–193. <https://doi.org/10.1016/j.envexpbot.2017.01.005>
- Sotero-Santos RB, Rocha O, Povinelli J (2005) Evaluation of water treatment sludges toxicity using the *Daphnia* bioassay. *Water Res* 39:3909–3917. <https://doi.org/10.1016/j.watres.2005.06.030>
- Takao TW, Bardini VS, de Jesus AD, Marchiori L, Albuquerque A, Fiore FA (2024) Beneficial use of water treatment sludge with stabilizers for application in road pavements. *Sustainability* 16:5333. <https://doi.org/10.3390/su16135333>
- Tamanini CR, Motta ACV, Andreoli CV, Doetzer BH (2008) Land reclamation recovery with the sewage sludge use. *Brazilian Arch Biol Technol* 51:843–855. <https://doi.org/10.1590/S1516-89132008000400023>
- Tan J, Huang Y, Chi B, Xiong Z, Zhou W, Yang Z, Zhou K (2024) Comparative study of iron and aluminium coagulants in conditioning sludge: sludge dewatering performance, physicochemical

- properties, and risk of heavy metal migration. *J Environ Chem Eng* 12:113168. <https://doi.org/10.1016/j.jece.2024.113168>
- Tytła M (2020) Identification of the chemical forms of heavy metals in municipal sewage sludge as a critical element of ecological risk assessment in terms of its agricultural or natural use. *Int J Environ Res Public Health* 17:1–22. <https://doi.org/10.3390/ijerph17134640>
- US EPA (2003) Ecological soil screening level for aluminum. Interim final, OSWER Directive 9285.7–60. U.S. Environmental Protection Agency, Office of Solid Waste and Emergency Response, Washington, D.C. Available online: <chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/> (accessed on February 17, 2026). https://www.epa.gov/sites/default/files/2015-09/documents/eco-sl_aluminum.pdf
- Weng L, Temminghoff EJM, Van Riemsdijk WH (2022) Aluminum speciation in natural waters: measurement using Donnan membrane technique and modeling using NICA-Donnan. *Wat Res* 36:4215–4226. [https://doi.org/10.1016/S0043-1354\(02\)00166-5](https://doi.org/10.1016/S0043-1354(02)00166-5)
- Yuan N, Pei Y, Bao A, Wang C (2020) The physiological and biochemical responses of *Daphnia Magna* to dewatered drinking water treatment residue. *Int J Environ Res Public Health* 17:5863. <https://doi.org/10.3390/ijerph17165863>
- Zhao Y, Liu R, Awe OW, Yang Y, Shen C (2018) Acceptability of land application of alum-based water treatment residuals – an explicit and comprehensive review. *Chem Eng J* 353:717–726. <https://doi.org/10.1016/j.cej.2018.07.143>
- Zloch J, Vavrková MD, Adamcová D, Radziemska M, Vyhnánek T, Trojan V, Đorđević B, Brtnický M (2018) Seasonal changes and toxic potency of landfill leachate for white mustard (*Sinapis alba* L.). *Acta Univ Agric Silvicae Mendelianae Brun* 66:235–242. <https://doi.org/10.11118/actaun201866010235>

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